Exercise J - A

1. The function f : [0, 3] → [1, 29], defined by f(x) = $2x^3 - 15x^2 + 36x + 1$, is : **[JEE-2012]**

(A) one-one and onto (B) onto but not one - one

(C) one-one but not onto (D) neither one- one nor onto

फलन f : [0, 3] → [1, 29] परिभाषित है, f(x) = $2x^3 - 15x^2 + 36x + 1$ तो f होगा : **[JEE-2012]**

(A) एकैकी लेकिन आच्छादक नहीं है (B) आच्छादक लेकिन एकैकी नहीं है

(C) एकैकी एवं आच्छादक दोनों है (D) न एकैकी न आच्छादक है

Ans. B

Sol. $f(x) = 2x^3 - 15x^2 + 36x + 1$

$\Rightarrow f'(x) = 6(x^2 - 5x + 6)$

$= 6(x - 2)(x - 3)$

$\because f(x)$ is non monotonic in $x \in [0,3]$

$\Rightarrow f(x)$ is not one-one

$f(x)$ is increasing in $x \in [0,2)$ and decreasing in $x \in (2,3]$

$f(0) = 1,\ f(2) = 29$ & $f(3) = 28$

$\therefore$ Range of $f(x)$ is $[1,29]$

$\Rightarrow f(x)$ is onto.

2. Let f : (–1, 1) → R be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then the value(s) of $f\left(\frac{1}{3}\right)$ is (are) : **[JEE-2012]**

माना f: (–1, 1) → R इस प्रकार की $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$, $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ तो $f\left(\frac{1}{3}\right)$ का मान होगा : **[JEE-2012]**

(A) $1 - \sqrt{\frac{3}{2}}$ (B) $1 + \sqrt{\frac{3}{2}}$ (C) $1 - \sqrt{\frac{2}{3}}$ (D) $1 + \sqrt{\frac{2}{3}}$

Ans. AB

Sol. $\because \theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \Rightarrow 2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$

Now $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$

$= \frac{1 + \cos 2\theta}{\cos 2\theta} = 1 + \frac{1}{\cos 2\theta}$ .....(i)

Let $\cos 4\theta = \frac{1}{3}$ $\Rightarrow$ $2\cos^2 2\theta - 1 = \frac{1}{3}$

$\Rightarrow$ $\cos 2\theta = \pm\sqrt{\frac{2}{3}}$

$\Rightarrow$ From (i), $f\left(\frac{1}{3}\right) = 1 \pm \sqrt{\frac{3}{2}}$

$\Rightarrow$ (A, B) are correct

3. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then : **[JEE(Adv.)-2014]**

(A) f(x) is an odd function
(B) f(x) is a one-one function
(C) f(x) is an onto function
(D) f(x) is an even function

माना कि $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ जहाँ $f(x) = (\log(\sec x + \tan x))^3$ के द्वारा परिभाषित किया गया है, तो : **[JEE(Adv.)-2014]**

(A) f(x) एक विषम फलन होगा
(B) f(x) एक ऐकैकी फलन होगा
(C) f(x) एक आच्छादक फलन होगा
(D) f(x) एक सम फलन होगा

Ans. ABC

Sol. (i) $f(-x) = -f(x)$ so it is odd function

(ii) $f'(x) = 3(\log(\sec x + \tan x))^2 \frac{1}{(\sec x + \tan x)} (\sec x \tan x + \sec^2 x) > 0$

(iii) Range of f(x) is R as $f\left(-\frac{\pi}{2}\right) \Rightarrow -\infty$ $\Rightarrow f\left(\frac{\pi}{2}\right) \Rightarrow \infty$

4. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is _____ **[JEE(Adv.)-2018]**

माना कि समुच्चय X में ठीक 5 अवयव हैं और समुच्चय Y में ठीक 7 अवयव हैं। यदि X से Y में एकैकी फलनों की संख्या α है और Y से X में आच्छादक फलनों की संख्या β है, तब $\frac{1}{5!}(\beta - \alpha)$ का मान है _____ **[JEE(Adv.)-2018]**

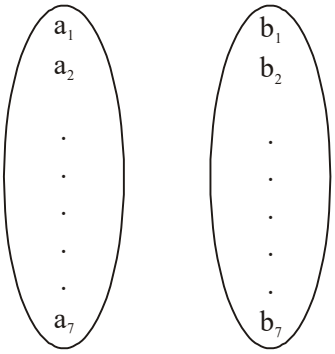
Ans. 119

Sol. $n(X) = 5$

$n(Y) = 7$

$\alpha \rightarrow$ Number of one-one function $= {}^7C_5 \times 5! = 21 \times 120 = 2520$

$\beta \rightarrow$ Number of onto function Y to X



1, 1, 1, 1, 3 1, 1, 1, 2, 2

$$\frac{7!}{3!4!} \times 5! + \frac{7!}{(2!)^3\, 3!} \times 5! = \left({}^7C_3 + 3.{}^7C_3\right)5! = 4 \times {}^7C_3 \times 5!$$

$$\frac{\beta - \alpha}{5!} = 4 \times {}^7C_3 - {}^7C_5 = 4 \times 35 - 21 = 119$$

5. If the function $f: R \rightarrow R$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE?

[JEE(Adv.)-2020]

(A) f is one-one, but NOT onto (B) f is onto, but NOT one-one

(C) f is BOTH one-one and onto (D) f is NEITHER one-one NOR onto

यदि फलन $f: R \rightarrow R$ को $f(x) = |x|(x - \sin x)$ से परिभाषित किया जाता है, तब निम्न में से कौनसा कथन सत्य है?

[JEE(Adv.)-2020]

(A) f एकैकी है, लेकिन आच्छादक नही है (B) f आच्छादक है, लेकिन एकैकी नहीं है

(C) f एकैकी एवं आच्छादक दोनों है (D) f एकैकी भी नहीं है एवं आच्छादक भी नहीं है

Ans. C

Sol. $f(x) \begin{cases} \rightarrow x(x - \sin x) & x \geq 0 \\ \rightarrow -x\,(x - \sin x) & x < 0 \end{cases}$

$\forall\ x \geq 0$

$f'(x) = 2x - (\sin x + x \cos x)$

$= (x - \sin x) + x\,(1 - \cos x) \geq 0$

$f(x)$ is increasing

$f(x))_{min} = f(0) = 0$

$x \rightarrow \infty \Rightarrow f(x) \rightarrow \infty$

$\Rightarrow \forall\ x \geq 0 \Rightarrow f(x) \in [0, \infty)$

$\forall\ x < 0$

$\because$ f is odd $\Rightarrow$ f(x) is also increasing

rangs of (x) for $x < 0$

$f(x) \in (-\infty\ , 0)$

Rangs of f(x) $=(-\infty, \infty) \Rightarrow$ f(x) is into

f(x) is increasing $\forall\ x \in R \Rightarrow$ f is one – one

6. Let the function $f: [0, 1] \to \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$. Then the value of

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) \text{ is.}$$ **[JEE(Adv.)-2020]**

माना कि फलन $f: [0, 1] \to \mathbb{R}$ को $f(x) = \frac{4^x}{4^x + 2}$ से परिभाषित किया जाता है। तब

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$$ का मान है। **[JEE(Adv.)-2020]**

Ans. 19

Sol. $f(x) = \frac{4^x}{4^x + 2}$; Replace $x \to 1 - x$

$$f(1-x) = \frac{4^{1-x}}{4^{1-x} + 2} = \frac{4 / 4^x}{4 / 4^x + 2} = \frac{2}{4^x + 2}$$

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{2}{4^x + 2} = 1\ \forall x.$$

So, $f\left(\frac{1}{40}\right) + f\left(\frac{39}{40}\right) = f\left(\frac{2}{40}\right) + f\left(\frac{38}{40}\right) = ... = f\left(\frac{19}{40}\right) + f\left(\frac{21}{40}\right) = 1$

$$\Rightarrow f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + ...f\left(\frac{19}{40}\right) + f\left(\frac{20}{40}\right) + f\left(\frac{21}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) = 19 + f\left(\frac{20}{40}\right) - f\left(\frac{1}{2}\right)$$

= 19

7. Let |M| denote the determinant of a square matrix. M. Let $g: \left[0, \frac{\pi}{2}\right] \to R$ be the function defined by

$$g(\theta) = \sqrt{f(\theta) - 1} + \sqrt{f\left(\frac{\pi}{2} - \theta\right) - 1}$$

$$\text{where } f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}.$$

Let p(x) be a quadratic polynomial whose roots are the maximum and minimum values of the function g(θ), and $p(2)=2-\sqrt{2}$. Then, which of the following is/are TRUE? **[JEE(Adv.)2022]**

माना |M|, एक वर्ग आव्यूह M के सारणिक को दर्शाता है। माना $g:\left[0,\frac{\pi}{2}\right]\to R$ एक फलन है जो

$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$ द्वारा परिभाषित है, जहाँ

$$\text{where } f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}$$ है।

माना p(x) एक द्विघात बहुपद है जिसके मूल, फलन g(θ) के अधिकतम तथा न्यूनतम मान हैं और $p(2)=2-\sqrt{2}$ है। तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं? **[JEE(Adv.)2022]**

(A) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$ (B) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$ (C) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$ (D) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

Ans. AC

Sol. $$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\frac{\pi}{4} & \tan\pi\end{vmatrix}$$

$$f(\theta)=\frac{1}{2}\begin{vmatrix}2 & \sin\theta & 1\\ 0 & 1 & \sin\theta\\ 0 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}0 & -\sin\left(\theta-\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & 0 & \log_e\left(\frac{4}{\pi}\right)\\ -\tan\left(\theta-\frac{\pi}{4}\right) & -\log_e\left(\frac{4}{\pi}\right) & 0\end{vmatrix}$$

$f(\theta)=(1+\sin^2\theta)+0$ (skew symmetric)

$$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$$

$=|\sin\theta|+|\cos\theta|$ for $\theta\in\left[0,\frac{\pi}{2}\right]$

$g(\theta)\in\left[1,\sqrt{2}\right]$

Again let $P(x)=k(x-\sqrt{2})(x-1)$

$2-\sqrt{2}=k(2-\sqrt{2})(2-1)$

$\Rightarrow k=1$ ($P(2)=2-\sqrt{2}$ given)

$\therefore P(x)=\left(x-\sqrt{2}\right)(x-1)$

for option (A) $P\left(\frac{3+\sqrt{2}}{4}\right)<0$ correct

option (B) $P\left(\frac{1+3\sqrt{2}}{4}\right)<0$ incorrect

option (C) $P\left(\frac{5\sqrt{2}-1}{4}\right)>0$ correct

option (D) $P\left(\frac{5-\sqrt{2}}{4}\right)>0$ incorrect

8. Let $f:R\to R$ be a function defined by $f(x)=\begin{cases}x^2\sin\left(\frac{\pi}{x^2}\right), & \text{if } x\neq 0,\\ 0, & \text{if } x=0\end{cases}$. Then which of the following statements is TRUE? **[JEE(Adv.)2024]**

(A) $f(x)=0$ has infinitely many solutions in the interval $\left[\frac{1}{10^{10}},\infty\right)$

(B) f(x) = 0 has no solutions in the interval $\left[\frac{1}{\pi},\infty\right)$

(C) The set of solutions of f(x) = 0 in the interval $\left(0,\frac{1}{10^{10}}\right)$ is finite

(D) f(x) = 0 has more than 25 solutions in the interval $\left(\frac{1}{\pi^2},\frac{1}{\pi}\right)$

माना कि $f : R \to R$ एक फलन (function) है, जो $f(x)=\begin{cases} x^2\sin\left(\frac{\pi}{x^2}\right), & \text{यदि } x \neq 0, \\ 0, & \text{यदि } x = 0 \end{cases}$ द्वारा परिभाषित है, तब निम्नलिखित कथनों में से कौनसा सत्य है? **[JEE(Adv.)2024]**

(A) अन्तराल (interval) $\left[\frac{1}{10^{10}},\infty\right)$ में f(x) = 0 के अनंत (infinitely many) हल (solutions) हैं

(B) अंतराल $\left[\frac{1}{\pi},\infty\right)$ में f(x) = 0 का कोई हल (solution) नहीं है

(C) अन्तराल $\left(0,\frac{1}{10^{10}}\right)$ में f(x) = 0 के हलों (solutions) का समुच्चय (set) परिमित (finite) है

(D) अंतराल $\left(\frac{1}{\pi^2},\frac{1}{\pi}\right)$ में f(x) = 0 के 25 से अधिक हल (solutions) हैं

Ans. D

Sol. for x > 0

$f(x) = 0 \Rightarrow x^2\sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \frac{\pi}{x^2} = n\pi$

$\Rightarrow x = \pm\frac{1}{\sqrt{n}}$

$x \in \left(\frac{1}{\pi^2},\frac{1}{\pi}\right) \Rightarrow \frac{1}{\pi^2} < \frac{1}{\sqrt{n}} < \frac{1}{\pi}$

$\pi < \sqrt{n} < \pi^2$

$\pi^2 < n < \pi^4$

f(x) = 0 has more than 25 solutions

9. Let f : R $\to$ R be a fuction such that f(x + y) = f(x) + f(y) for all x, y $\in$ R, and g : R $\to$ (0, ∞) be a function such that g(x + y) = g(x) g(y) for all x, y $\in$ R. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is _____. **[JEE(Adv.)2024]**

माना कि f : R $\to$ R एक ऐसा फलन (fuction) है कि सभी x, y $\in$ R के लिए f(x + y) = f(x) + f(y) है, और g : R $\to$ (0, ∞) एक ऐसा फलन है कि सभी x, y $\in$ R के लिए g(x + y) = g(x) g(y) है। यदि $f\left(\frac{-3}{5}\right) = 12$ और $g\left(\frac{-1}{3}\right) = 2$ हैं, तब $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 51

Sol. f : R $\to$ R

f(x + y) = f(x) + f(y) for all x, y $\in$ R

$\Rightarrow$ f(x) = ax

Now $f\left(\frac{-3}{5}\right) = 12$

a = – 20

$\therefore$ f(x) = – 20x

Again, g (x + y) = g(x). g(y) x, y $\in$ R.

$\Rightarrow$ f(x) = k^x

Now $g\left(\frac{-1}{3}\right) = 2$

$2 = k^{\frac{-1}{3}}$

$\Rightarrow \left(2^{-1}\right)^3 = k$

$$k = \frac{1}{8}$$

$$\therefore\ g(x) = \left(\frac{1}{8}\right)^x$$

Now, $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right) g(0)$

$\Rightarrow [-5 + 64 - 8] \cdot 1 \quad \Rightarrow 64 - 13 \qquad \Rightarrow 51$

PARAGRAPH

Paragraph for Q. No 10 to 11

Let S = {1, 2, 3, 4, 5, 6} and X be the set of all relations R from S to S that satisfy both the following properties:

i. R has exactly 6 elements.

ii. For each (a, b) $\in$ R, we have $|a - b| \geq 2$.

Let Y = {R $\in$ X : The range of R has exactly one element } and

Z = {R $\in$ X : R is a function from S to S}.

Let n(A) denote the number of elements in a set A.

(There are two questions based on PARAGRAPH)

माना कि S = {1, 2, 3, 4, 5, 6} है, और X, S से S में उन सभी संबंधों (relations) R का समुच्चय (set) है जो निम्नलिखित दोनों गुणधर्मों (properties) को संतुष्ट करते हैं :

i. R में ठीक (exactly) 6 अवयव (elements) हैं।

ii. प्रत्येक (a, b) $\in$ R के लिए $|a - b| \geq 2$ है।

माना कि Y = {R $\in$ X : R के परिसर (range) में ठीक (exactly) एक अवयव (element) है} और

Z = {R $\in$ X : R, S से S में एक फलन (function) है।}

माना कि n(A), समुच्चय A में अवयवों की संख्या (number of elements) को दर्शाता है।

(PARAGRAPH पर आधारित दो प्रश्न हैं, नीचे दिया गया प्रश्न उनमें से एक है)

10. If $n(X) = {}^mC_6$, then the value of m is _____. **[JEE(Adv.)2024]**

यदि $n(X) = {}^mC_6$ है, तब m का मान _____ है। **[JEE(Adv.)2024]**

Ans. 20

11. If the value of $n(Y) + n(Z)$ is k^2, then $|k|$ is _____. **[JEE(Adv.)2024]**

यदि $n(Y) + n(Z)$ का मान k^2 है, तब $|k|$ _____ है। **[JEE(Adv.)2024]**

Ans. 36

Sol. All possible ordered pair (a, b) s.t. $|a - b| \geq 2$

(1, 3) (2, 4) (3, 1) (4, 1) (5, 1) (6, 1)

(1, 4) (2, 5) (3, 5) (4, 2) (5, 2) (6, 2)

(1, 5) (2, 6) (3, 6) (4, 6) (5, 3) (6, 3)

(1, 6) (6, 4)

10. $n(X) = {}^{20}C_6 \Rightarrow m = 20$

11. $n(Y) = 0$ as maximum 4 ordered pairs are possible having same image

$n(Z) = 4 \times 3 \times 3 \times 3 \times 3 \times 4 = 1296$

$n(Y) + n(Z) = 1296 = (36)^2$

$\Rightarrow |k| = 36$

12. Let R denote the set of all real numbers. Let $a_i, b_i \in R$ for $i \in \{1,2,3\}$.

Define the functions $f : R \to R$, $g : R \to R$, and $h : R \to R$ by

$f(x) = a_1 + 10x + a_2x^2 + a_3x^3 + x^4$,

$g(x) = b_1 + 3x + b_2x^2 + b_3x^3 + x^4$,

$h(x) = f(x + 1) - g(x + 2)$.

If $f(x) \neq g(x)$ for every $x \in R$, then the coefficient of x^3 in $h(x)$ is : **[JEE(Adv.)2025]**

मान लीजिए कि R सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है। मान लीजिए कि $i \in \{1,2,3\}$ के लिए $a_i, b_i \in R$ हैं। फलनों (functions) $f : R \to R$, $g : R \to R$, और $h : R \to R$ को

$f(x) = a_1 + 10x + a_2x^2 + a_3x^3 + x^4$,

$g(x) = b_1 + 3x + b_2x^2 + b_3x^3 + x^4$,

$h(x) = f(x + 1) - g(x + 2)$.

द्वारा परिभाषित कीजिए। यदि प्रत्येक $x \in R$ के लिए $f(x) \neq g(x)$ है, तब $h(x)$ में x^3 का गुणांक (coefficient) है :

[JEE(Adv.)2025]

(A) 8 (B) 2 (C) –4 (D) –6

Ans. C

Sol. Equation $f(x) = g(x)$ will not have any real solution.

$(a_3 - b_3)x^3 + (a_2 - b_2)x^2 + 7x + (a_1 - b_1) = 0$

$a_3 = b_3$ & $49 - 4(a_2 - b_2)(a_1 - b_1) < 0$

$h(x) = a_1 + 10(x + 1) + a_2(x + 1)^2 + a_3(x + 1)^3 + (x + 1)^4$

$-b_1 - 3(x + 2) - b_2(x + 2)^2 - b_3(x + 2)^3 - (x + 2)^4$

On RHS only $(x + 1)^4 - (x + 2)^4$ will have term containing x^3

So, coefficient of $x^3 = {}^4C_1 - {}^4C_1(2)$

$= 4 - 8 = -4$

13. Let N denote the set of all natural numbers, and Z denote the set of all integers. Consider the functions $f: N \to Z$ and $g: Z \to N$ defined by

$$f(n) = \begin{cases} \frac{(n+1)}{2} & \text{if n is odd,} \\ \frac{(4-n)}{2} & \text{if n is even,} \end{cases}$$

and

$$f(n) = \begin{cases} 3+2n & \text{if} \quad n \geq 0 \\ -2n & \text{if} \quad n < 0 \end{cases}$$

Define $(g \cdot f)(n) = g(f(n))$ for all $n \in N$, and $(f \cdot g)(n) = f(g(n))$ for all $n \in Z$.

Then which of the following statements is (are) TRUE ? **[JEE(Adv.)2025]**

(A) $g \cdot f$ is **NOT** one-one and $g \cdot f$ is **NOT** onto

(B) $f \cdot g$ is **NOT** one-one but $f \cdot g$ is onto

(C) g is one-one and g is onto

(D) f is **NOT** but f is onto

मान लीजिए कि N सभी प्राकृत संख्याओं के समुच्चय को दर्शाता है, और Z सभी पूर्णांकों के समुच्चय को दर्शाता है। फलनों $f: N \to Z$ और $g: Z \to N$ जो

$$f(n) = \begin{cases} \frac{(n+1)}{2} & \text{य दि n विषम है,} \\ \frac{(4-n)}{2} & \text{य दि n सम है,} \end{cases}$$

और

$$f(n) = \begin{cases} 3+2n & \text{य दि} \quad n \geq 0 \text{ है} \\ -2n & \text{य दि} \quad n < 0 \text{ है} \end{cases}$$

द्वारा परिभाषित हैं, पर विचार कीजिए। सभी $n \in N$ के लिए $(g \cdot f)(n) = g(f(n))$, और सभी $n \in Z$ के लिए $(f \cdot g)(n) = f(g(n))$ परिभाषित कीजिये।

तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) ? **[JEE(Adv.)2025]**

(A) $g \cdot f$ कएकैकी **नहीं** है और $g \cdot f$ आच्छादक भी **नहीं** है

(B) $f \cdot g$ एकैकी **नहीं** है परंतु $f \cdot g$ आच्छादक है

(C) g एकैकी है और g आच्छादक भी है

(D) f एकैकी **नहीं** है परंतु f आच्छादक है

Ans. AD

Sol. $f(n)=\begin{cases}\dfrac{n+1}{2} & n \to \text{odd} \\ \dfrac{4-n}{2} & n \to \text{even}\end{cases}$ $\qquad N \to Z$

$f(1) = 1$ $\qquad$ $f(5) = 3$

$f(2) = 1$ $\qquad$ $f(6) = -1$ $\qquad$ not one-one and onto

$f(3) = 2$ $\qquad$ $f(7) = 4$

$f(4) = 0$ $\qquad$ $f(8) = -2$

$g(n)=\begin{cases}3+2n, & n \geq 0 \\ -2n, & n < 0\end{cases}$

$g(0) = 3$ $\qquad$ $g(-1) = 2$

$g(1) = 5$ $\qquad$ $g(-2) = 4$ $\qquad$ not one-one and into

$g(2) = 7$ $\qquad$ $g(-3) = 6$

$g(3) = 9$ $\qquad$ $g(-4) = 8$

fog(x)

$-4 \Rightarrow -2$

$-3 \Rightarrow -1$

$-2 \Rightarrow 0$

$-1 \Rightarrow 1$ $\qquad$ one-one and onto

$0 \Rightarrow 2$

$1 \Rightarrow 3$

$2 \Rightarrow 4$

$3 \Rightarrow 5$

gof(x)

$1 \Rightarrow 5$

$2 \Rightarrow 5$

$3 \Rightarrow 7$

$4 \Rightarrow 3$

$5 \Rightarrow 9$

$6 \Rightarrow 2$

$7 \Rightarrow 11$

$8 \Rightarrow 4$

not one-one and into

14. Let R denote the set of all real numbers. Let $f : R \to R$ be a function such that $f(x) > 0$ for all $x \in R$, and $f(x + y) = f(x) f(y)$ for all $x, y \in R$.

Let the real numbers $a_1, a_2, ..., a_{50}$ be in an arithmetic progression. If $f(a_{31}) = 64f(a_{25})$, and $\sum_{i=1}^{50} f(a_i) = 3\left(2^{25}+1\right)$,

then the value of $\sum_{i=6}^{30} f(a_i)$ is _____. **[JEE(Adv.)2025]**

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है। मान लीजिए कि $f : R \to R$ एक इस प्रकार का फलन है कि सभी $x \in R$ के लिए $f(x) > 0$ है, और सभी $x, y \in R$ के लिए $f(x + y) = f(x) f(y)$ है।

मान लीजिए कि वास्तविक संख्याएँ $a_1, a_2, ..., a_{50}$ एक समांतर श्रेणी में हैं। यदि $f(a_{31}) = 64f(a_{25})$ है, और $\sum_{i=1}^{50} f(a_i) = 3\left(2^{25}+1\right)$,

है, तब $\sum_{i=6}^{30} f(a_i)$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. 96

Sol. $f : R \to R$

$f(x) > 0 \quad \forall x \in R$

$f(x + y) = f(x).f(y) \ \forall x, y \in R$

$\Rightarrow f(x) = a^{\lambda x}$; $\lambda \in R$

Now $a_1, a_2, \ldots\ldots\ldots a_{50} \rightarrow$ A.P.

$f(a_{31}) = 64\ f(a_{25})$

$a^{\lambda(A + 30D)} = 64\ a^{\lambda(A + 24\ D)}$

$a^{\lambda A} . .a^{30\lambda D} = 64\ a^{\lambda A} . a^{\lambda 24D}$

$a^{6\lambda D} = 64$ .....(1)

Again $6a^{6\lambda D} = 2^6 \Rightarrow a = 2$; $\lambda D = 1$

$f(a_1) + f(a_2) + f(a_3) + \ldots\ldots\ldots + f(a_{50}) = 3(2^{25} + 1)$

$a^{\lambda A} + a^{\lambda(A + D)} + a^{\lambda(A + 2D)} + \ldots\ldots\ldots + a^{\lambda(A + 49D)} = 3(2^{25} + 1)$

$a^{\lambda A} (1 + a^{\lambda D} + a^{2\lambda D} + \ldots\ldots\ldots + a^{49\lambda D}) = 3(2^{25} + 1)$

$$a^{\lambda A}\left\{\frac{1-\left(a^{\lambda D}\right)^{50}}{1-a^{\lambda D}}\right\} = 3\left(2^{25}+1\right)$$

$$2^{\lambda A}\left\{\frac{1-(2)^{50}}{1-2}\right\} = 3\left(2^{25}+1\right)$$

$$2^{\lambda A}\left\{\left(2^{25}-1\right)\left(2^{25}+1\right)\right\} = 3\left(2^{25}+1\right)$$

$2^{\lambda A} (2^{25} - 1) = 3$

$$2^{\lambda A} = \frac{3}{2^{25}-1}$$

$$\sum_{i=6}^{30} f\left(a_i\right) = f\left(a_6\right) + f\left(a_7\right) + \ldots + f\left(a_{30}\right)$$

$= a^{\lambda(A + 5D)} + a^{\lambda(A + 6D)} + \ldots\ldots\ldots + a^{\lambda(A + 29D)}$

$= a^{\lambda A}\ \{a^{5\lambda D} + a^{6\lambda D} + \ldots\ldots\ldots + a^{29\lambda D}\}$

$= 2^{\lambda A}\ \{2^5 + 2^6 + \ldots\ldots\ldots + 2^{29}\}$

$$\Rightarrow 2^{\lambda A}\left\{\frac{2^5\left(1-2^{25}\right)}{1-2}\right\}$$

$$\frac{3}{2^{25}-1} \times \frac{2^5\left(2^{25}-1\right)}{1}$$

$= 32 \times 3$

$= 96$

15. Let S = {1, 2, 3, ..., 10}. Consider the set X = {R : R is an equivalence relation on the set S such that R has exactly 42 elements}. Then the number of elements in X is _____. **[JEE(Adv.)2026]**

मान लीजिए कि S = {1, 2, 3, ..., 10} है। समुच्चय X = {R : R, समुच्चय S पर एक ऐसा तुल्यता संबंध है कि R में ठीक 42 अवयव हैं } पर विचार कीजिए। तब X में अवयवों की संख्या _____है। **[JEE(Adv.)2026]**

Ans. 2520

Sol. A relation R on set S = {1, 2, ..., 10} has exactly 42 ordered pairs (a, b).

1. Reflexivity: For every element $a \in S$, the pair $(a, a) \in R$.

   Since there are 10 elements in S, all 10 diagonal pairs– (1,1), (2,2), ..., (10,10) must be in R.

   Remaining elements to choose = 42 – 10 = 32

2. Symmetry: If $(a, b) \in R$ (where $a \neq b$), then (b, a) must also be in R.

   This means the remaining 32 elements must come in pairs.

   Number of symmetric pairs (a, b) with $a \neq b = \frac{32}{2} = 16$

3. Transitivity: If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

Concurrently with symmetry, this means if any set of elements are related to one another, every single element in that subset must be related to every other element in that same subset. They form closed, mutually related groups.

If a mutually related group has x elements, it will contribute exactly x^2 ordered pairs to the relation R (since every element pairs with all x elements, including itself).

Let the 10 distinct elements of S be divided into k mutually related groups, with sizes $x_1, x_2, ..., x_k$.

1. Total number of elements: $x_1 + x_2 + ... + x_k = 10$

2. Total number of ordered pairs: $x_1^2 + x_2^2 + \ldots + x_k^2 = 42$

Case I: Group sizes of 6, 2, 1, 1 $\Rightarrow (6^2 + 2^2 + 1^2 + 1^2 = 36 + 4 + 1 + 1 = 42)$

Case II: Group sizes of 5, 4, 1 $\Rightarrow (5^2 + 4^2 + 1^2 = 25 + 16 + 1 = 42)$

the number of ways to divide n distinct objects into groups of sizes p, q, r, ... is given by the formula:

$$\text{Number of ways} = \frac{n!}{(p! \cdot q! \cdot r! \ldots) \times a! \cdot b! \ldots}$$

(where a, b, ... represent the number of times a group size repeats).

Calculation for Case I: Group sizes (6, 2, 1, 1)

$$\text{Ways}_1 = \frac{10!}{6!\cdot 2!\cdot 1!\cdot 1!\times 2!} \Rightarrow \quad \text{Ways}_1 = \frac{10\times 9\times 8\times 7\times 6!}{6!\times 2\times 1\times 2} = \frac{5040}{4} = 1260$$

Calculation for Case II: Group sizes (5, 4, 1)

$$\text{Ways}_2 = \frac{10!}{5!\cdot 4!\cdot 1!} \Rightarrow \quad \text{Ways}_2 = \frac{10\times 9\times 8\times 7\times 6\times 5!}{5!\times 24} = \frac{30240}{24} = 1260$$

Total number of relations = $\text{Ways}_1 + \text{Ways}_2$

Total = 1260 + 1260 = 2520

16. Let N denote the set of all positive integers. Consider the sets

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{1, 2, 3, 4, 5, 6, 7\}$$

Let S be the set of all functions $f: A \to B$ such that $f(2) \neq 2$ and $f(4) \neq 4$. Consider the set $T = \{f \in S$: there exists a function $g: B \to N$ such that $g(f(x)) = 2^x$ for all $x \in A\}$. Then the number of elements in the set T is _____.

[JEE(Adv.)2026]

मान लीजिए कि N सभी धन पूर्णांकों के समुच्चय को दर्शाता है। समुच्चयों

$$A = \{1, 2, 3, 4, 5\} \text{ और } B = \{1, 2, 3, 4, 5, 6, 7\}$$

पर विचार कीजिए। मान लीजिए कि S उन सभी फलनों $f: A \to B$, जिनके लिए $f(2) \neq 2$ और $f(4) \neq 4$ हैं, का समुच्चय है। समुच्चय $T = \{f \in S$: एक ऐसे फलन $g: B \to N$ का अस्तित्व है कि सभी $x \in A$ के लिए $g(f(x)) = 2^x$ है$\}$ पर विचार कीजिए।

तब समुच्चय T में अवयवों की संख्या _____ है। **[JEE(Adv.)2026]**

Ans. 1860

Sol. $g(f(x)) = 2^x \ \forall\ x \in A$

because 2^x is one-one function.

So, f(x) must be one-one function.

So, total no. of one-one function = $7 \times 6 \times 5 \times 4 \times 3 = 2520$

but $f(2) \neq 2$ and $f(4) \neq 4$

So, required function = Total – (when f(2) = 2 OR f(4) = 4) + (when f(2) = 2 and f(4) = 4)

$= 2520 - ({}^6C_4 \times 4! + {}^6C_4 \times 4!) + {}^5C_3 \times 3!$

$= 2520 - (360 + 360) + 60$

$= 1860$